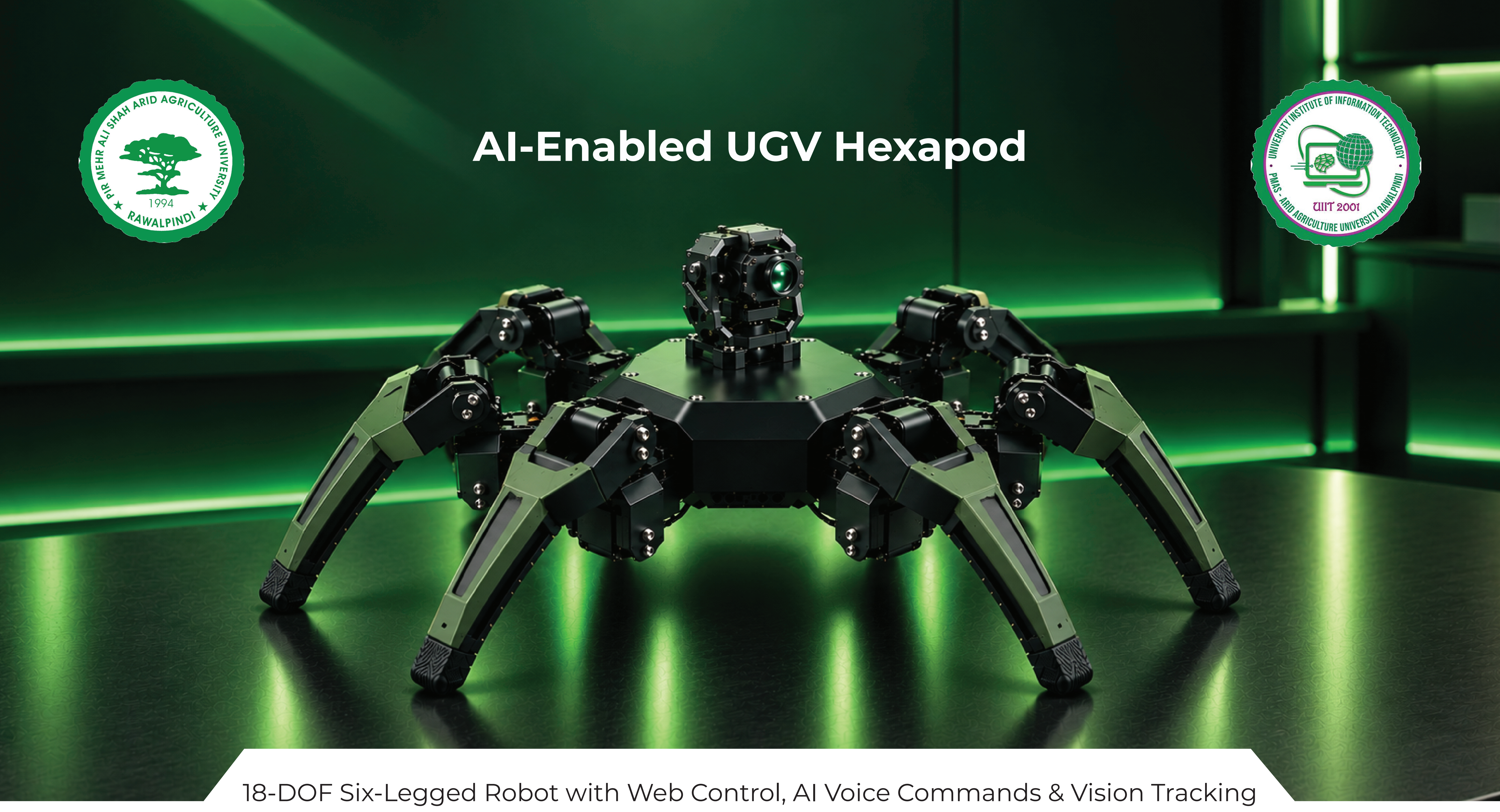
PIR MEHR ALI SHAH ARID AGRICULTURE UNIVERSITY 1994 RAWALPINDI

AI-Enabled UGV Hexapod

UNIVERSITY INSTITUTE OF INFORMATION TECHNOLOGY PMAS-ARID AGRICULTURE UNIVERSITY RAWALPINDI UIIT 2001



18-DOF Six-Legged Robot with Web Control, AI Voice Commands & Vision Tracking

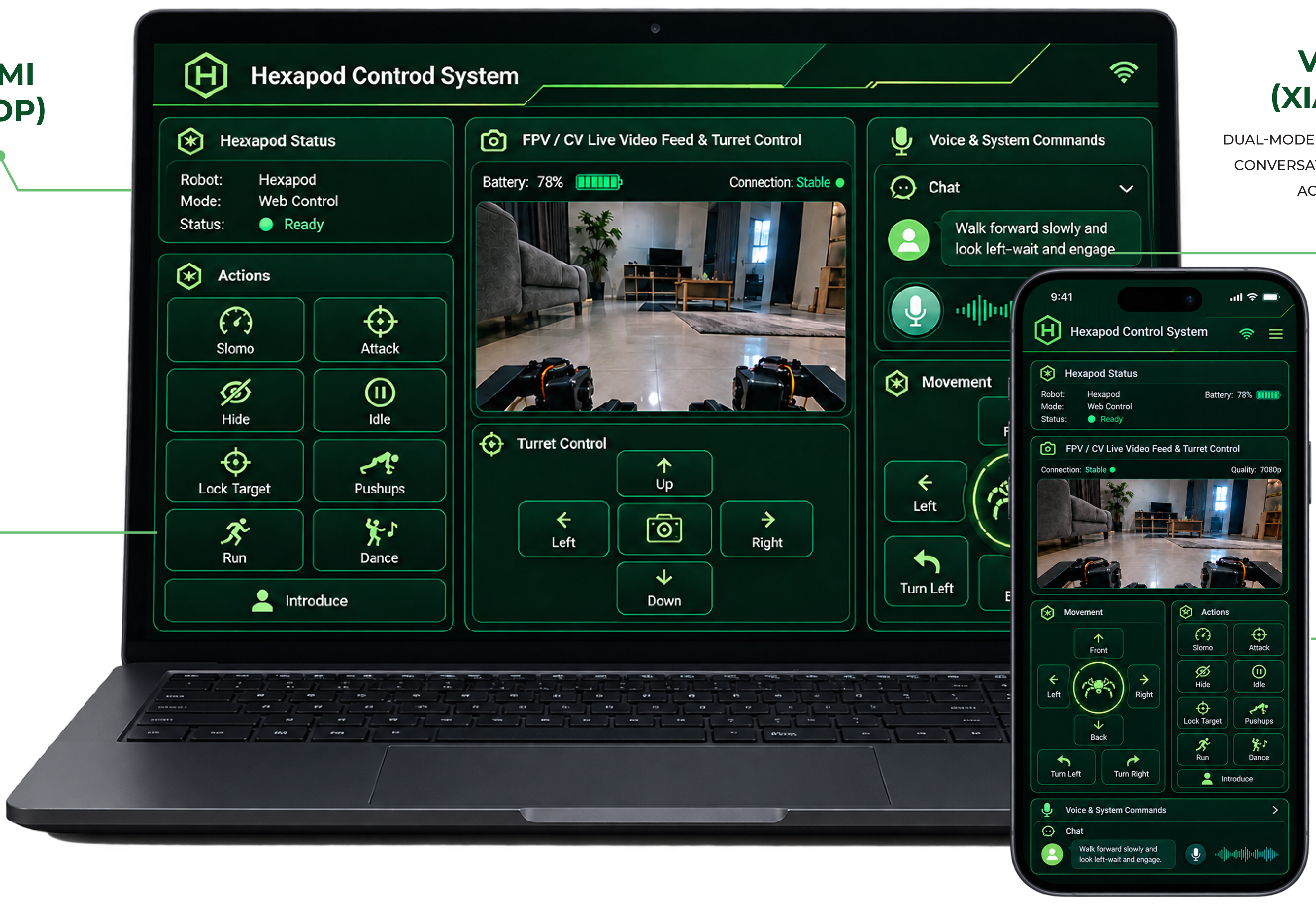
INTRODUCTION

Inspired by nature's six-legged movement, the AI-Enabled UGV Hexapod is a research platform for exploring legged robotics, inverse kinematics, ROS 2, simulation, and embedded AI. It combines six legs with web-based control, AI voice commands, a movable live-view camera, and person/object tracking. It also supports additional sensors, cameras, and task-specific equipment for different research and field applications.

PROBLEM STATEMENT

Different environments require different robotic solutions. Drones can fly over varied terrain but face challenges in low visibility and confined spaces, while wheeled and tracked robots may struggle with stairs, gaps, and rubble. This project explores a legged ground robot for navigating challenging surfaces and inspecting hazardous areas remotely.

SYSTEM OVERVIEW



Robot: Hexapod
Mode: Web Control
Status: Ready

Battery: 78%
Connection: Stable
Quality: 7080p

Walk forward slowly and look left-wait and engage

Hexapod Controd System
Hexapod Control System
9:41
Hexapod Status
FPV / CV Live Video Feed & Turret Control
Voice & System Commands
Chat
Actions
Slomo
Attack
Hide
Idle
Lock Target
Pushups
Run
Dance
Introduce
Turret Control
Up
Left
Right
Down
Movement
Front
Back
Turn Left
Turn Right

WEB HMI (DESKTOP)

VOICE AI (XIAOZHI AI)

DUAL-MODE INTERACTION SUPPORTING CONVERSATIONAL CHAT AND DIRECT ACTION COMMANDS

PREDEFINED ACTIONS

TRIGGER INSTANT MOTION ROUTINES LIKE SLOMO, ATTACK, HIDE, PUSHUPS, AND DANCE.

WEB HMI (MOBILE)

OBJECTIVES & SCOPE

To explore nature-inspired six-legged movement and its ability to navigate uneven terrain and obstacles.

To provide a hands-on platform for learning and experimenting with inverse kinematics, gait planning, ROS 2, and robotics simulation.

To control the robot using a web interface or voice commands.

To investigate vision-based surveillance and remote inspection through live video, person detection, and object tracking.

TOOLS & TECHNOLOGIES

Fusion 360
3D Model / PCB Design

Illustrator
Design

URDF/Xacro
Robot model

ROS 2 Humble
Middleware

Python
Gait / IK / (OpenCV)

C++
Firmware code

Gazebo
Physics sim

Ubuntu Linux
OS

GitHub
Repository

CSS3
Styling

HTML5
Structure

JavaScript
Interface logic

PROJECT MEMBERS

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SUPERVISED BY

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